Business Requirements Document (BRD)

Apple Product Portfolio Performance Optimization

1. Document Control

Attribute	Details
Document Title	Business Requirements Document (BRD) – Apple Product Portfolio Analysis
Project Name	Apple Product Portfolio Performance Optimization
Prepared By	Senior Data Analyst / Business Analyst
Reviewed By	TBC (To Be Confirmed)
Version	1.0
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Classification	Internal

2. Project Background

Business Background: In highly competitive global electronics and consumer retail markets, understanding product performance across diverse categories (Smartphones, Laptops, Tablets, Wearables, and Accessories) is critical to maintaining market leadership, optimizing inventory, and maximizing revenue generation.

Existing Business Problems: The business currently lacks automated visibility into pricing trends, consumer sentiment patterns, and resource distribution across product lines. Pricing strategies are highly reactive, and without clear data segmentation, it is difficult to determine which product configurations (combinations of CPU, RAM, and storage) yield the highest market satisfaction and performance margins.

Why This Analysis is Needed: By leveraging an expansive portfolio dataset containing 100k distinct device configurations, the business can shift from a reactive strategy to data-driven proactive positioning. This analysis will identify pricing optimization windows, highlight poor-performing product archetypes, and uncover technical specifications heavily favored by consumers to optimize the production pipeline.

3. Business Objectives

- **Portfolio Optimization:** Identify top-performing and under-performing product configurations to refine product lifecycle management.

- **Strategic Pricing Strategy:** Uncover correlations between launch years, specific component variants (RAM, CPU), and price distribution to guide competitive retail pricing.
- **Quality & Customer Satisfaction Alignment:** Analyze product rating patterns across core categories to mitigate customer dissatisfaction risks.

4. Business Questions This Analysis Must Answer

1. Which core product categories generate the highest average customer ratings, and which are under-performing?
2. How have launch prices for individual product lines shifted year-over-year from 2015 to 2024?
3. What is the impact of technical specifications (e.g., M-series chips vs. A-series Bionic chips) on retail pricing and final customer ratings?
4. Do high-tier storage variants (e.g., 1TB/2TB) justify their pricing premiums based on corresponding user satisfaction metrics?
5. How is the 100k portfolio volume currently distributed across core product segments (iPhone, iPad, MacBook, Watch, AirPods), and is the product mix optimized for consumer demand?

5. Project Scope

In Scope:

- Exploratory Data Analysis (EDA) to map distributions of price, ratings, and launch years.
- Data cleaning, preparation, and structure profiling of the 100k records dataset.
- Calculation of localized portfolio metrics and key performance indicators.
- Production-grade dashboard delivery summarizing performance, category mixes, and technical component metrics.

Out of Scope:

- Integration with real-time lived transactional enterprise resource planning (ERP) systems.
- External macroeconomic factors (e.g., inflation rates, supply chain component constraints, competitive market pricing indices).
- Predictive real-time price adjustment automation engines.

6. Data Sources

Detail	Source Metadata
Dataset Name	Apple Products Dataset — 100K Rows & 20 Features
Source	Kaggle (apple_products_dataset_100k.csv)
Dataset Size	13.43 MB | 100,000 Records | 20 Columns
Time Period Covered	Launch Years spanning 2015 to 2024

Important Columns Monitored:

- **product_id:** Unique identifier assigned to each product entry.
- **category:** Broad classification (iPhone, iPad, MacBook, Apple Watch, AirPods, etc.).
- **model_name:** Specific designation tag (e.g., Model-754, Model-214).
- **release_year:** The calendar year the specific configuration model was launched.
- **price:** Retail/launch evaluation price expressed in USD.
- **rating:** Average consumer customer evaluation score (Scaled 1.0 – 5.0).
- **color:** External aesthetic layout variation.
- **storage / ram / cpu:** Primary technical hardware component specifications.

Data Limitations and Quality Concerns:

- **Synthetic Nature:** The dataset is synthetically generated and may contain distributions or hardware combinations that deviate slightly from actual real-world production configurations.
- **Missing Sales Volumetrics:** Actual total market units sold per row are missing; analysis must focus heavily on catalog configuration distributions and customer ratings.

7. Key Definitions and KPIs

KPI Name	Definition	Business Formula	Business Interpretation
Portfolio Average Rating (PAR)	Overall customer satisfaction baseline across a chosen segment.	SUM(rating) / COUNT(product_id)	Determines user adoption and satisfaction health.
Premium Configuration Index (PCI)	The share of products positioned within the top-tier pricing boundaries.	(COUNT(price > $2500) / TOTAL products) * 100	Indicates structural layout exposure toward high-end lines.
Category Distribution Mix (CDM)	The balance ratio of active configurations within the structural catalog.	(COUNT(Category X) / TOTAL products) * 100	Highlights potential portfolio oversaturation or area gaps.

8. Analytical Framework / Business Segmentation

- **Tiered Price Architecture Segmentation:** High, Medium, and Budget tier buckets based on launch price distribution percentiles to evaluate profit opportunities.
- **Hardware Capability Matrix:** Grouping configurations by computational power (e.g., M-Series vs. A-Series processors coupled with RAM allocations) to trace product efficiency and positioning trends.
- **Sentiment Quadrant Mapping:** Cross-referencing average price positions with product ratings to identify 'Value Premium' options versus 'At-Risk Line items'.

9. Deliverables

- **Cleaned Dataset:** Exportable repository script documenting type conversions and anomaly handling.
- **SQL Analysis:** Executable script containing window transformations and category aggregates.
- **Python Analysis:** Jupyter Notebook covering deep Exploratory Data Analysis (EDA) and distribution graphs.
- **KPI Calculations:** Dedicated analytical ledger containing mapped calculations segmented by launch year.
- **Power BI Dashboard:** Secure, dynamic enterprise visualization report tracking product configuration trends.
- **Business Insights and Recommendations:** Executive presentation summary outlining clear portfolio roadmap recommendations.

10. Success Criteria

- **Execution Coverage:** 100% of the 100,000 internal configuration rows are verified, handled, and correctly accounted for.
- **Stakeholder Acceptance:** Executive stakeholders confirm that reports effectively address all core portfolio business queries.

11. Stakeholders

Role	Responsibility
Product Portfolio Manager	Primary consumer of the analysis; utilizes output metrics to determine upcoming design configurations and lifecycle status.
Pricing Strategy Committee	Reviews historical multi-year pricing models across categories to guide launch value benchmarks for future hardware tiers.
Senior Data Analyst	Responsible for technical data cleansing, transformation script execution, validation testing, and dashboard design.

12. Assumptions and Constraints

- **Assumptions:** Data structures, column names, and field definitions will remain static throughout the execution timeframe.
- **Data Limitations:** The synthetic nature of the file may result in edge-case asset configurations. Analysts must implement robust logic sorting filters to handle anomalous data points safely.
- **Technical Constraints:** Processing must be completed using standard local desktop analytics environments (SQL, Python, Excel, Power BI).
- **Project Dependencies:** Delivery schedules depend on secure data ingestion pipelines and server access provisions for the analytical workspace tools.